Mapping & Analyzing Participation in Tutor/Mentor Leadership and Networking Conferences

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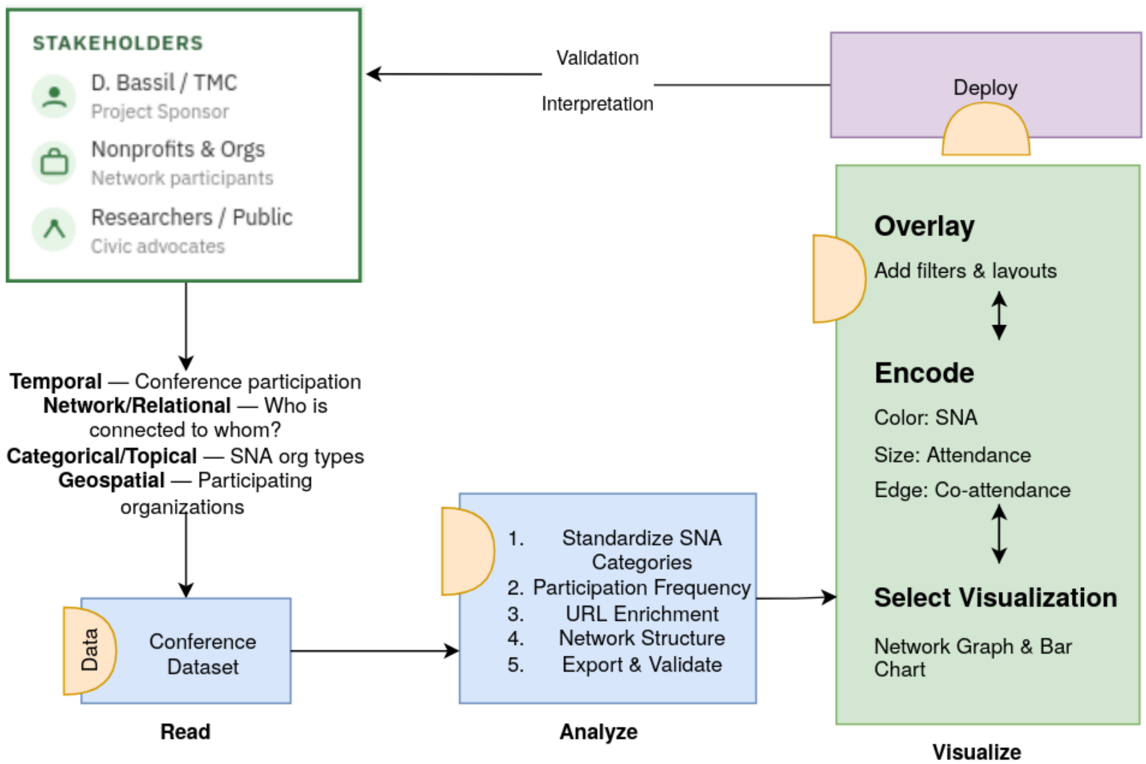

Fig. 1. Visual abstract of the project pipeline using the Börner et al. (2018) visualization framework. The diagram is organized around four phases, Read, Analyze, Visualize, and Deploy, with a feedback loop connecting deployment outcomes back to interpretation and re-analysis.

1. INTRODUCTION AND PRIOR WORK

This project builds on work completed by Fall 2025 student teams who created an open-source template for collecting event participation data and generating network analysis maps from Tutor/Mentor Conference data. The next phase will improve the Kumu and network-analysis tools by making the sector-based graph modes fully functional, with consistent data schemas, stronger validation, and reliable exports for platforms such as Gephi and Kumu. The project will also strengthen the analytics dashboard by adding network metrics, distribution views, top-entity tables, interactive filters, and data-quality indicators that help users better interpret the data before export. Its practical value is in making long-term participation patterns easier to explore, compare, and understand while helping users connect event history, organizational relationships, and current Tutor/Mentor resources.

1.1 Stakeholder Groups

This project serves four primary stakeholder groups. First, the project sponsor (Daniel F. Bassil, Tutor/Mentor Connection) requires an updated, publicly accessible Kumu.io visualization that communicates the conference network's history and allows him to share the tool with partners and funders. His priorities are visual clarity, organizational accuracy, and the inclusion of website links to current program information. Second, conference participants and their organizations — including youth-serving nonprofits, businesses, colleges, K–12 schools, government agencies, faith communities, and foundations — benefit from a map that documents their historical involvement and connects them with similar organizations. Third, researchers and civic advocates studying nonprofit collaboration and community networks require a dataset that is well-documented, consistently categorized, and accessible for further analysis. Fourth, members of the general public and prospective partners visiting tutormentorexchange.net need an intuitive, visually engaging interface to understand the scope of the network and find organizations they may wish to connect with.

1.2 Stakeholder Needs

The project sponsor needs the Kumu map to support three specific enhancements: (1) color coding of participant nodes by organization type (Program, Business, College, K–12 School, Government, Faith, Foundation, etc.) so users can visually distinguish stakeholder categories at a glance; (2) a search-by-participant feature so that any individual or organization can find which conferences they attended; and (3) hyperlinks from each organization node to its website, drawing on the directory at tutormentorexchange.net/chicago-area-program-links.

Conference participants and their organizations need the map to accurately represent their type and involvement history, with correct organizational names and active website links. Researchers need the underlying dataset to be well-structured, consistently labeled, and documented for reproducibility. The general public needs the visualization to be self-explanatory, with a descriptive narrative panel explaining what the map shows and how to navigate it. Across all groups, the shared need is for a visualization that is publicly accessible, visually clear, and regularly maintainable by future student teams.

2. DATA ACQUISITION

Data for this project was sourced primarily from the cleaned conference participation spreadsheet prepared by the Fall 2025 TeamK student group, which itself was derived from original Excel records maintained by Daniel F. Bassil documenting attendance at Tutor/Mentor Leadership and Networking Conferences from May 1994 through November 2015. The dataset was shared by the client via Google Sheets and has undergone two rounds of student-led cleaning. Secondary data will be collected by cross-referencing participant organizations against the Chicago-area program directory maintained at tutormentorexchange.net/chicago-area-program-links to populate missing website URLs. No personally identifiable information beyond professional affiliations and conference attendance is included in the dataset.

2.1 Data Sources

The primary data source is the TeamK cleaned conference participation dataset (updated February 11, 2026). This spreadsheet consolidates records originally maintained by the client in Excel across multiple conference years. A secondary data source is the online program directory at tutormentorexchange.net/chicago-area-program-links, which will be used to look up and append website URLs for participating organizations. The original unprocessed conference spreadsheets are archived by the client and were used as the basis for both the 2025 TeamK cleaning effort and this project.

2.2 Data Description, Quality and Coverage

The dataset contains 6,428 participation records across approximately 42 conference events spanning May 1994 through November 2015 (conferences were held twice yearly, in May and November, with a small number of gaps). Each record represents one individual's attendance at a specific conference and includes 18 fields: UniqueID, Salutation, First Name, Last Name, SNA Category (original and cleaned), Title, Title (cleaned), Organization (original and cleaned), Active/Inactive status, Address, City, State, Zip, Conference date, Month, and Year. The SNA Category (Social Network Analysis category) field classifies each participant by organization type. The cleaned category distribution includes: Program (~3,205 records, ~50%), Resource (~608), College (~438), Other (~382), Intermediary (~186), Business (~128), T/MC (~125), Government (~124), K–12 School (~340 across case variants), Faith (~118), and Foundation (~55). Some inconsistency remains in category casing (e.g., "program" vs. "Program", "college" vs. "College") that Timothy will standardize as part of the data cleaning phase. Geographic coverage is predominantly Chicago and the surrounding metropolitan area, with a small number of out-of-state participants. The dataset has no missing UniqueID values and relatively complete name fields, though a significant portion of organization name and title fields remain blank for many records, which may limit the depth of organizational-level analysis.

3. DATA ANALYSIS

[Primary planning, to be updated when received] The planned analyses for this project are as follows. First, participant type standardization: the SNA Category field will be normalized to a consistent controlled vocabulary (Program, Business, College, K–12 School, Government, Faith, Foundation, Intermediary, T/MC, Resource, Other) using Python (Pandas) to resolve casing inconsistencies and map ambiguous entries. Second, participation frequency analysis: a summary will be computed showing how many unique conferences each participant attended, which years had the highest attendance, and how participation rates varied by organization type over the 1994–2015 period. Third, organization website enrichment: a lookup process will cross-reference participant organization names against the tutormentorexchange.net directory to append URL data. This will be performed semi-manually with assistance from scripted string matching. Fourth, network structure analysis: the cleaned and enriched dataset will be formatted for import into Kumu.io, structuring participants as nodes and shared conference attendance as edges to enable network-level exploration.

4. VISUALIZATIONS

The primary visualization for this project is an enhanced Kumu.io network participation map. The map will display each conference participant as a node and their conference attendance as connections, enabling users to explore the structure of the Tutor/Mentor network over time. The planned enhancements, color coding by participant type, search-by-participant functionality, and embedded website links, will be updated by Patrick's hand sketches (see **Fig. 2**) when received. The existing Kumu.io map serves as the baseline visualization that this project will build upon. Additional planned visualizations include a participation frequency chart showing attendance trends over time by organization type, which will help stakeholders identify which sectors have been most consistently engaged.

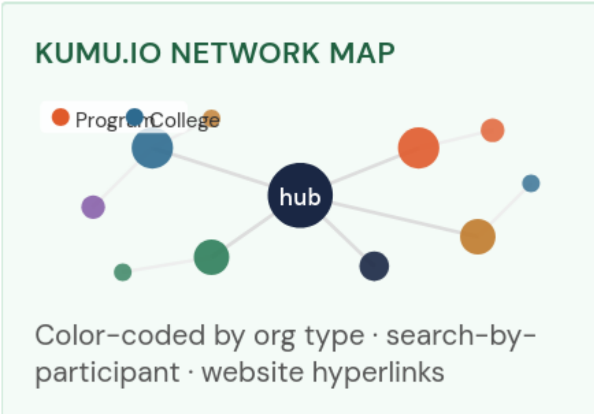

Figure 2. A sketch of the updated Kumu.io visualization that implements color coding.

4.1 Kumu.io Network Participation Map

The primary deliverable is an updated Kumu.io network map (**Fig. 2**) built on the Fall 2025 TeamK template. Each participant is represented as a node, colored by their SNA organization type. Nodes will include embedded hyperlinks to organizational websites where available. The map will support a search-by-participant function so any user can filter the view to see exactly which conferences a specific person or organization attended. The existing baseline can be explored at tutormentorexchange.net/mapping-participation.

4.2 Participation Frequency Visualization

[Sketch to be updated when received.] A secondary planned visualization (**Fig. 3**) is a participation frequency chart showing the count of attendees per conference event over time, segmented by organization type. This chart will allow the client and stakeholders to quickly see which years had peak participation, which types of organizations have been most consistently engaged, and where gaps in outreach may have occurred. The chart will be implemented using a Python visualization library (e.g., Matplotlib or Plotly) and exported for inclusion in both the report and the Kumu.io supporting page.

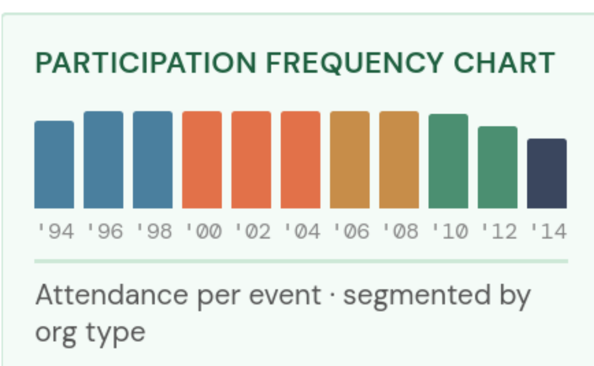

Figure 3. Participation frequency chart that can reflect commitment and will have color coding that reflects **Fig. 2**.

5. USAGE AND CRITIQUE OF AI TOOLS

AI tools have already played a role in this project's planning and documentation phases. Claude (Anthropic) was used to help structure the team's activity plan and organize project tasks in Trello. For data cleaning, AI-assisted string matching may be used to semi-automate the lookup of organization website URLs from the tutormentorexchange.net directory. In the visualization phase, LLM-assisted code generation may be used to accelerate Kumu.io configuration scripting. Limitations of these tools include potential hallucinations in factual content (requiring human verification), inability to access live or proprietary data without manual input, and limitations in understanding the specific social context of Chicago-area nonprofit networks. The team will document all AI tool usage and manually verify AI-generated content before including it in final deliverables.

6. INTERPRETATION OF RESULTS

At this stage of the project, analysis and visualization are planned but not yet executed. Upon completion, this section will address the following research questions: (1) Which organization types have been most consistently represented at Tutor/Mentor conferences over time, and how has this distribution changed between 1994 and 2015? (2) Which participants or organizations attended the greatest number of events, and what does this suggest about the core of the network? (3) Are there geographic or organizational patterns in conference attendance that could inform future outreach strategy? The interpretation will be grounded in the client's mission of expanding tutor/mentor program reach across Chicago's high-poverty neighborhoods, and results will be communicated directly to Daniel F. Bassil for review and recognition as the source of this work, in line with his stated publication notes.

ACKNOWLEDGEMENTS

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